WSPRadar.org reconstruction & comparison

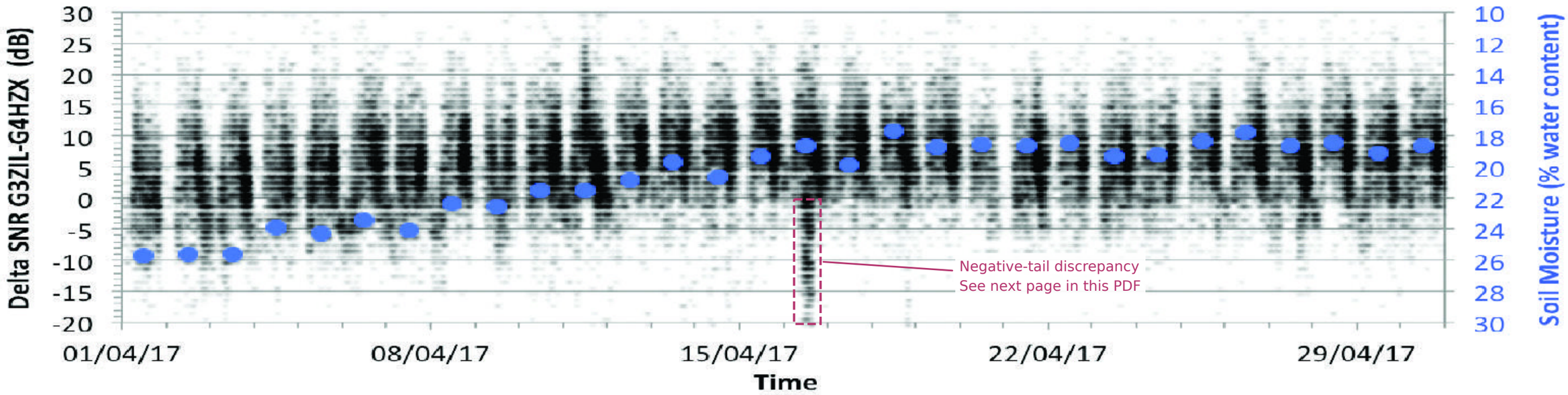
SNR differences over time

Referenced publication: Gwyn Griffiths (G3ZIL) and Nigel Squibb (G4HZX) (2017). *Improving HF Band SNR.*

Source figure: **Figure 3, p. 24** · Practical Wireless, October 2017, pp. 23-26

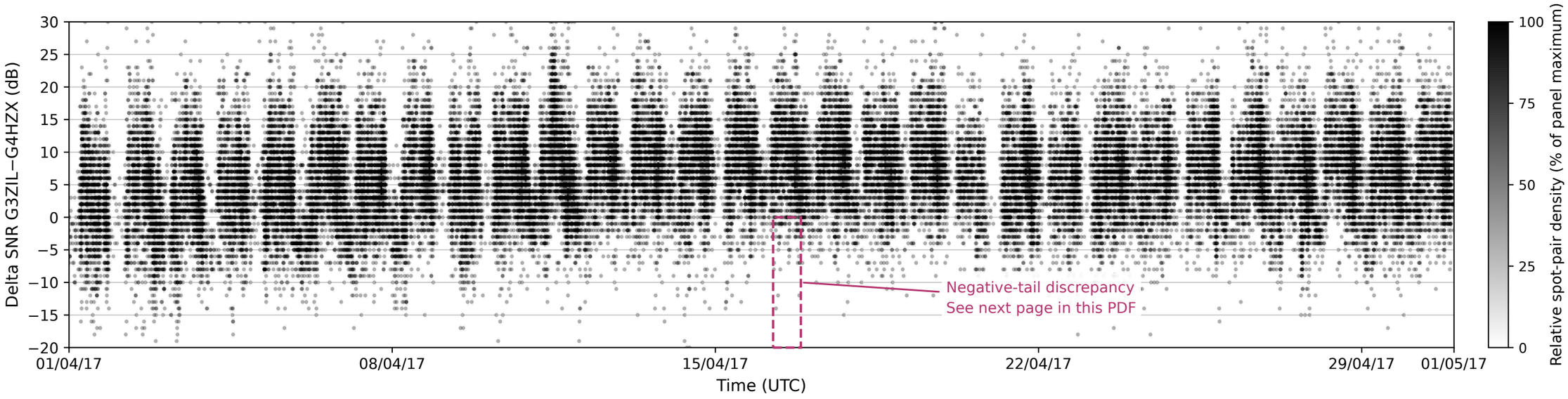
Demo: RX Benchmark | April 2017 | 40 m | ΔSNR = Target G3ZIL (IO90HW) − Reference G4HZX (IO91) | 57 767 spot pairs | Reference correction 0.0 dB

A Original image from publication

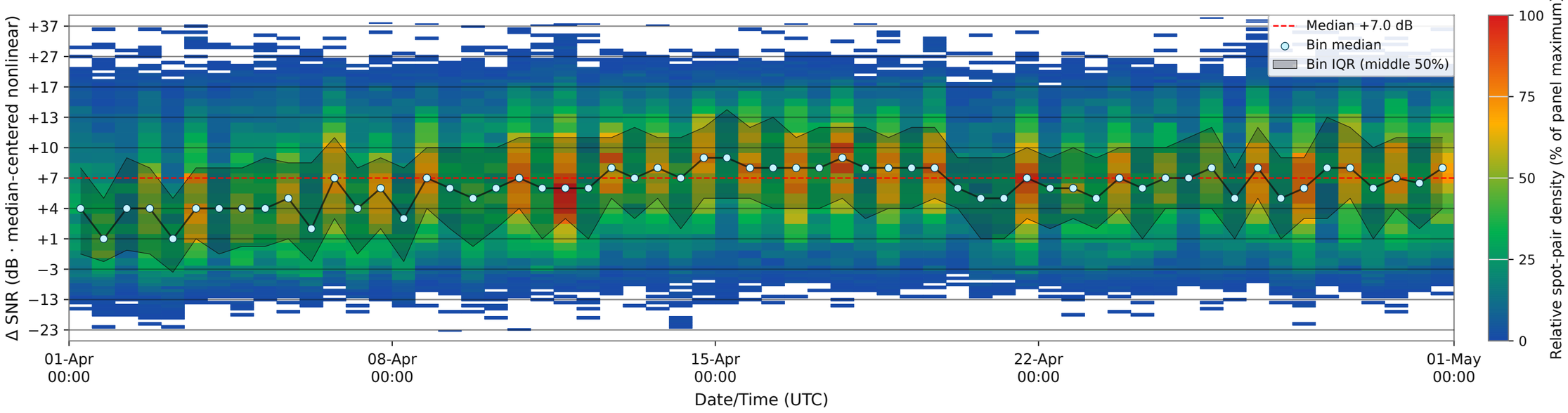


B Reconstruction from WSPRadar data

All 57 767 pairs; 1h × 1dB density



C WSPRadar view · Δ SNR over Time · 12h bins



B: All paired dots over 1h density; the scale describes the background. C: 12h bins, medians, middle 50% (IQR) and nonlinear axis.
WSPRadar reconstruction settings: min_qualifying_stations_per_map_segment = 1; min_spot_pairs_per_station = 1.
min_decode_event_share_percent = 1.0 (ΔDecode maps only; does not filter these ΔSNR plots). The demo opens with 24h bins; C uses 12h bins.

WSPRadar.org reconstruction & comparison

Pairing and duplicate reports

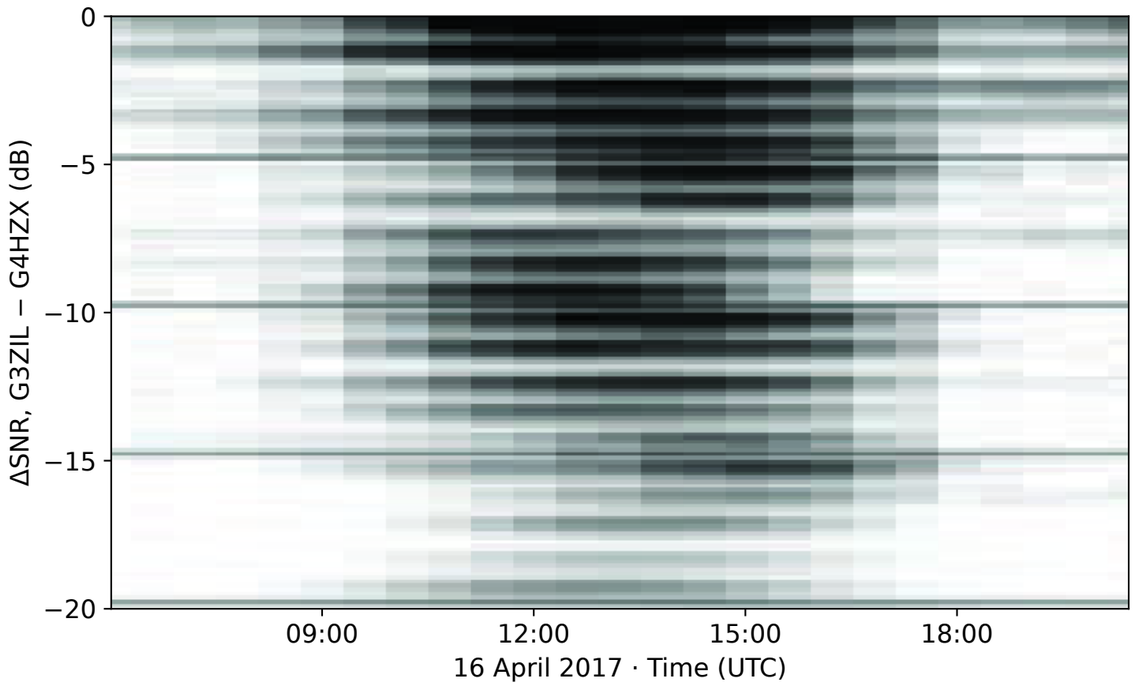
Referenced publication: Gwyn Griffiths (G3ZIL) and Nigel Squibb (G4HZX) (2017). *Improving HF Band SNR.*

Source figure: **Figure 3, p. 24** · Practical Wireless, October 2017, pp. 23-26

Demo: RX Benchmark | 16 April 2017 diagnostic | 40 m | ΔSNR = Target G3ZIL (IO90HW) − Reference G4HZX (IO91) | Reference correction 0.0 dB | Displayed range: 0 to −20 dB

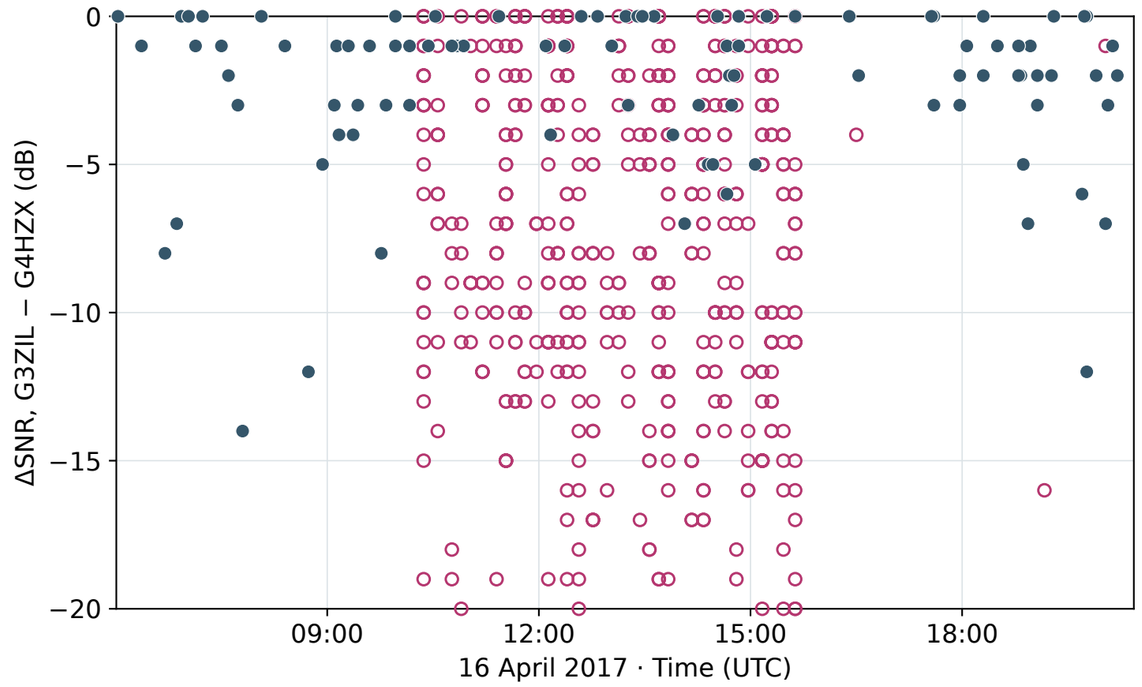
Enlarged detail

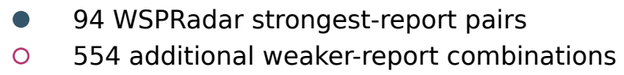
A Original image from publication



Strongest pairs and additional combinations

B Database reports





A receiver can report the same transmitter more than once within one WSPR cycle, at different reported frequencies. Pairing every report from one receiver with every report from the other creates additional SNR differences. WSPRadar retains the strongest qualifying report at each receiver and calculates one paired difference.

Worked example: one transmitting station → G3ZIL · 16 April 2017, 10:46 UTC

Reported frequency (MHz)	Reported SNR (dB)
7.040194	−27
7.040125	−15
7.040094	**−5**
7.040033	−26

The table lists four G3ZIL reports of this station. For the same transmitter and cycle, G4HZX supplied two further reports: −18 and −8 dB (not shown). Pairing each table row with each G4HZX report produces 4 × 2 = 8 possible SNR differences. WSPRadar uses the strongest report at each receiver:
G3ZIL − G4HZX = −5 − (−8) = +3 dB.
This retained pair is above the plot's upper limit of 0 dB; five of the other seven combinations fall inside it. All six reports specify 43 dBm TX power, so power normalization does not change these differences.

What explains the April 16 concentration?

One transmitting callsign accounts for 551 of the 554 additional weaker-report combinations in the displayed window. The callsign is available upon request. These combinations do not represent separate transmissions.

Evidence from other receivers

At 10:46 UTC, both comparison receivers and many other stations reported a weaker signal from this transmitter approximately 30-31 Hz above their strongest report, typically about 10 dB lower in SNR. The wider database shows the pattern becoming widespread at 03:22 UTC and declining sharply at 15:46 UTC, even as the strongest reports became stronger. Matching frequency offsets and relative strengths across receivers favour additional spectral components associated with the transmitted signal. Possible causes include unwanted modulation in the transmitter's oscillator, power supply or audio chain [1]. The physical cause remains unconfirmed; database reports alone cannot exclude a shared decoding artifact.

Effect on the monthly comparison

Multiple reports affect 211 of 57 767 selected paired cycles (0.365%) across April. Excluding this one callsign leaves the monthly median ΔSNR unchanged at +7 dB. This supports the stability of the monthly median, but does not establish that every transmission was free of unwanted spectral components.

The 648 combinations cover only the displayed range and time window (approximately 06:01-20:26 UTC). The all-combinations calculation can explain the negative plume; the authors' exact pairing procedure is not known.

Additional analysis of April 2017 WSPR database reports, retrieved 26 September 2026. Frequency and SNR values were checked directly. Original figure: p. 24, Figure 3. [1] Griffiths, Elmore and Robinett, Some Observations While Using the KiwiSDR to Spot WSPR Stations, sections 4 and 7.